

Mathematics(P&C)- TMH

Time Allowed : 1 hour _____ **Maximum Marks : 50**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. Section I contains 5 questions. Each question in this section contains 3 marks. 1 mark will be deducted for a wrong response.
2. Section II contains 5 question. The question carries 5 marks. 2 marks will be deducted for a wrong response.
3. Section II contains 2 subjective type questions. Each question carries 5 marks.

SECTION – I

This section contains 5 questions. Only one option is correct.

- Q1:** The number of integral solutions of $x + y + z = 0$ with $x \geq -5$, $y \geq -5$, $z \geq -5$ is
- A) 134
 - B) 136
 - C) 138
 - D) 140
- Q2:** $2m$ white counters and $2n$ red counters are arranged in a straight line with $(m+n)$ counters on each side of a central mark. The number of ways of arranging the counters, so that the arrangements are symmetrical with respect to the central mark, is
- A) ${}^{m+n}C_m$
 - B) ${}^{2m+2n}C_{2m}$
 - C) $\frac{1}{2} \frac{(m+n)!}{m!n!}$
 - D) None of these
- Q3:** Ten different letters of an alphabet are given. Words with five letters are formed from these given letters. The number of words which have at least one of the letters repeated is
- A) 69760
 - B) 30340
 - C) 99748
 - D) None of these
- Q4:** The number of squares which can be formed on a chessboard is
- A) 160
 - B) 224
 - C) 204
 - D) None of these
- Q5:** The number of zeros at the end of $100!$ is
- A) 20
 - B) 22
 - C) 24
 - D) 26

SECTION – II

This section contains 1 question. More than one options may be correct

- Q6:** Letters of the word SUDESH can be arranged in
- A) 120 ways when two vowels are together
 - B) 180 ways when vowels occur in alphabetical order
 - C) 24 ways when vowels and consonants occupy their respective place
 - D) 240 ways when vowels do not occur together
- Q7:** On Diwali, all the students of a class send greeting cards to one another. If the postmen deliver 1640 greeting cards to the students of this class, then the number of students in the class is.
- A) 40
 - B) 41
 - C) a prime
 - D) None of these
- Q8:** A box contains two white, three black and four red balls. The number of ways in which we can select three balls from the box, if at least one black ball is to be included in the selection, is
- A) a perfect square
 - B) 2^5
 - C) divisible by 7
 - D) None of these
- Q9:** A seven digit number is divisible by 3 is to be formed using 7 out of numbers 1, 2, 3, ..., 9. The number of ways in which this can be done is
- A) $12(7!)$
 - B) $7(8!)$
 - C) $2(8!)$
 - D) None of these
- Q10:** If n objects are arranged in a row, then the number of ways of selecting three of these objects so that no two of them are next to each other is
- A) $\frac{1}{6}(n-2)(n-3)(n-4)$
 - B) ${}^{n-2}C_3$
 - C) ${}^{n-3}C_3 + {}^{n-3}C_2$
 - D) None of these

SECTION – III

This section contains 2 Subjective questions which have to be solved in the space provided.

- Q11:** Eighteen guests have to be seated, half on each side of a long table. Four particular guests desire to sit on one particular side and three others on the other side. Determine the number of ways in which the sitting arrangements can be made.

Q12: A committee of 12 is to be formed from nine women and eight men. In how many ways can this be done if at least five women have to be included in a committee? In how many of these committees

- a) the women are in majority ?
- b) the men are in majority ?